MẠCH ĐIỆN TỬ 3

60 TIẾT

NỘI DUNG MÔN HỌC

- Chương 1 Mạch Lọc
- Chương 2 Khuếch Đại công suất cao tần và nhân tần
- Chương 3 Mạch Dao động và tổng hợp tần số
- Chương 4 Điều chế tương tự
- Chương 5 Máy Phát và máy thu

Tài liệu tham khảo:

[1] Giáo trình Điện tử thông tin - Phạm Hồng Liên , Nhà xuất bản ĐHQG TP. HCM

[2] Mạch điện tử 3 - Mạch điện tử thông tin , Hoàng Đình Chiến , Nhà xuất bản ĐHQG TP. HCM

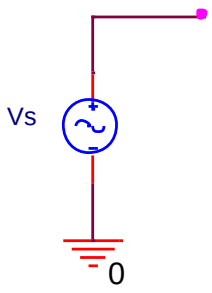
[3] Electronic communication technology , James K. Hardy, Prentice-Hall

[4] RF Microelectronics , Behzad Razavi , Prentice-Hall

Chương 1 Mạch Lọc

- Mạch lọc thụ động
  - Mạch lọc thông thấp, thông cao RC bậc 1
  - Mạch lọc RLC bậc 2
- Mạch lọc tích cực
  - Mạch lọc tích cực bậc 1
  - Mạch lọc tích cực bậc 2
  - Mạch lọc thông dải

Giới thiệu


Vs
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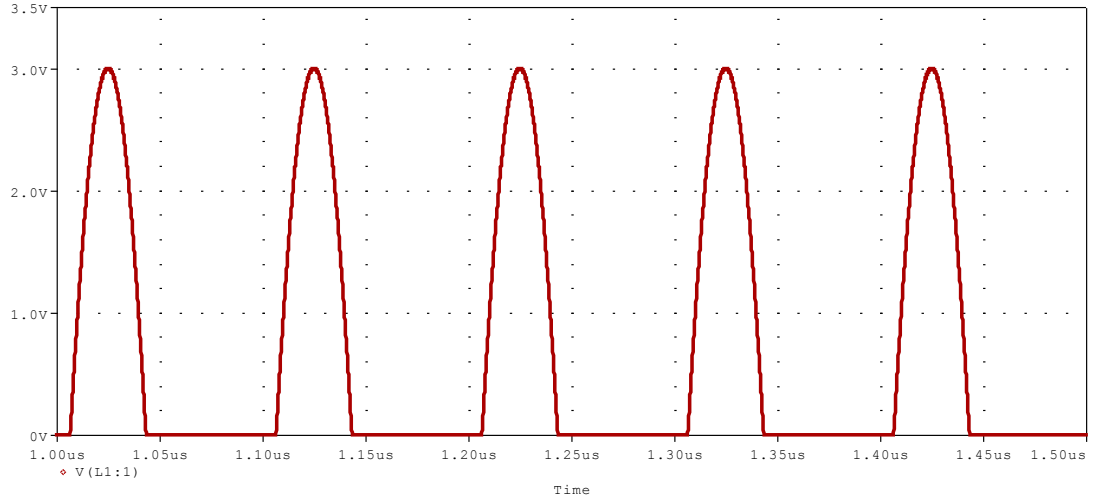
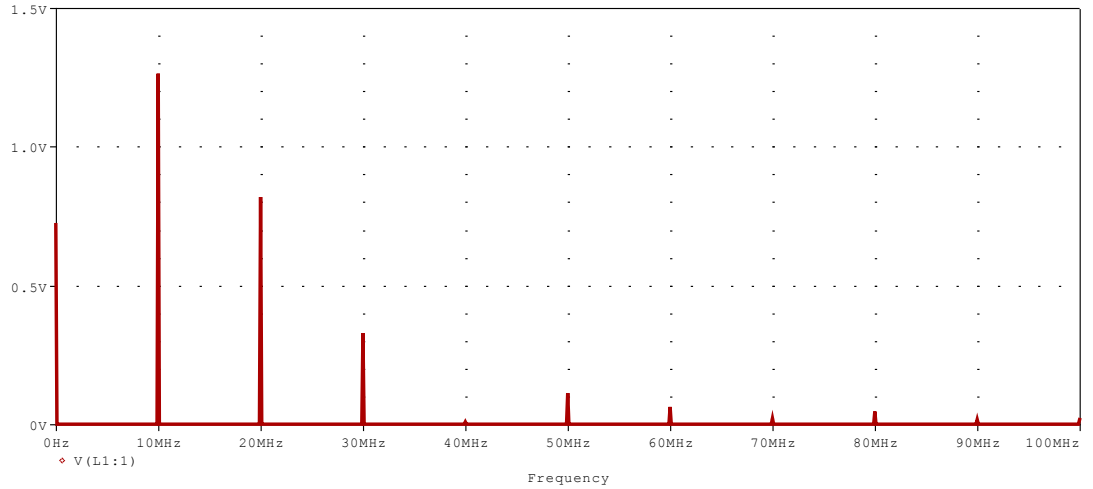
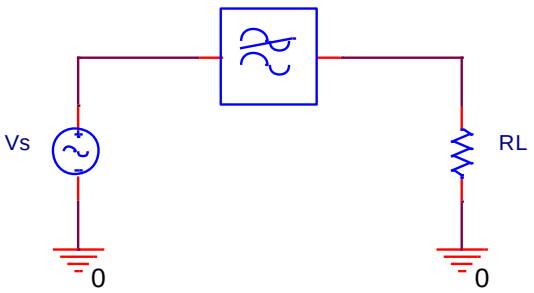
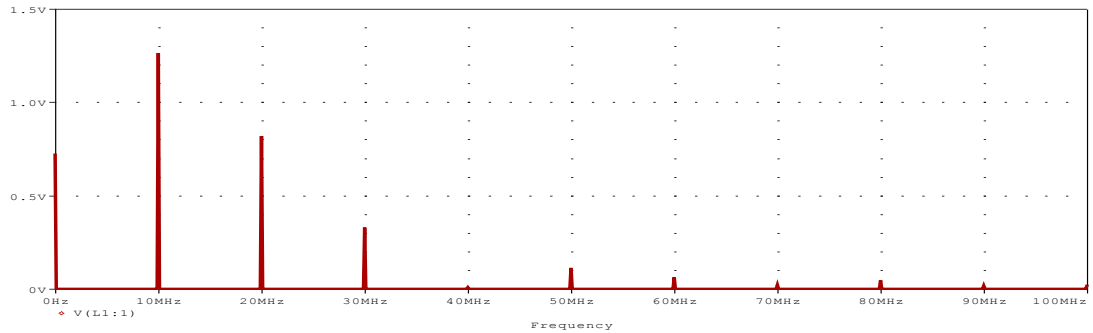
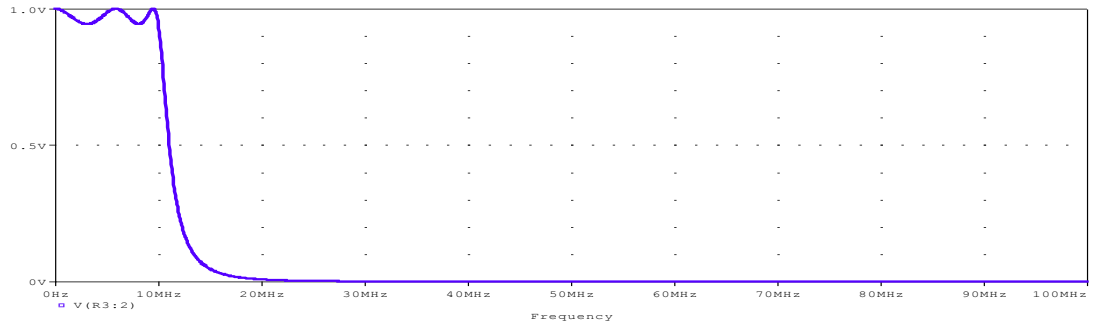
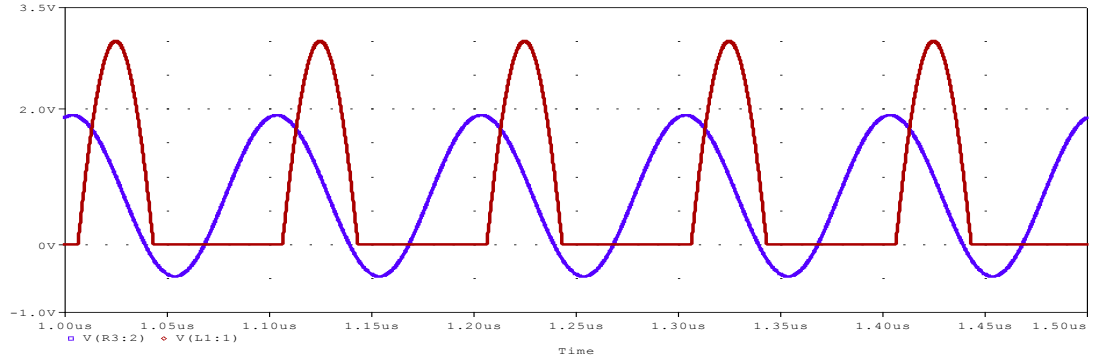

3.5V
3.0V
2.0V
1.0V
0V
1.00us
1.05us
1.10us
1.15us
1.20us
1.25us
1.30us
1.35us
1.40us
1.45us
1.50us
V(L1:1)
Time

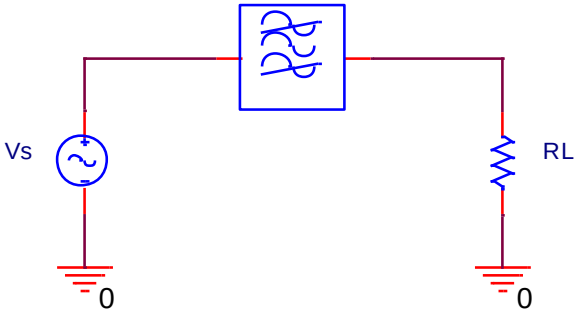

1.5V
1.0V
0.5V
0V
0Hz
10MHz
20MHz
30MHz
40MHz
50MHz
60MHz
70MHz
80MHz
90MHz
100MHz
V(L1:1)
Frequency

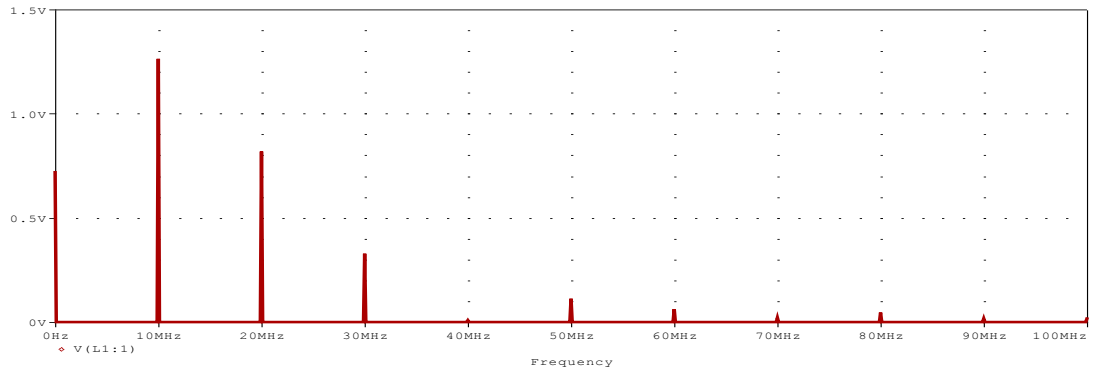

Vs
RL
0
0

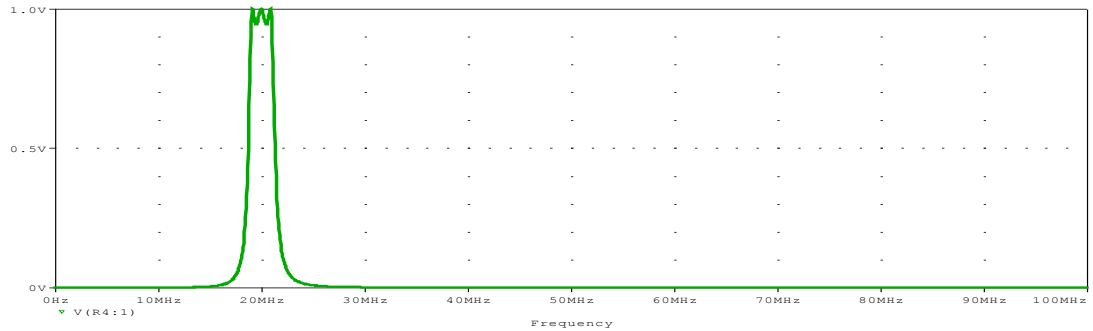

1.5V
1.0V
0.5V
0V
0Hz
10MHz
20MHz
30MHz
40MHz
50MHz
60MHz
70MHz
80MHz
90MHz
100MHz
V(L1:1)
Frequency

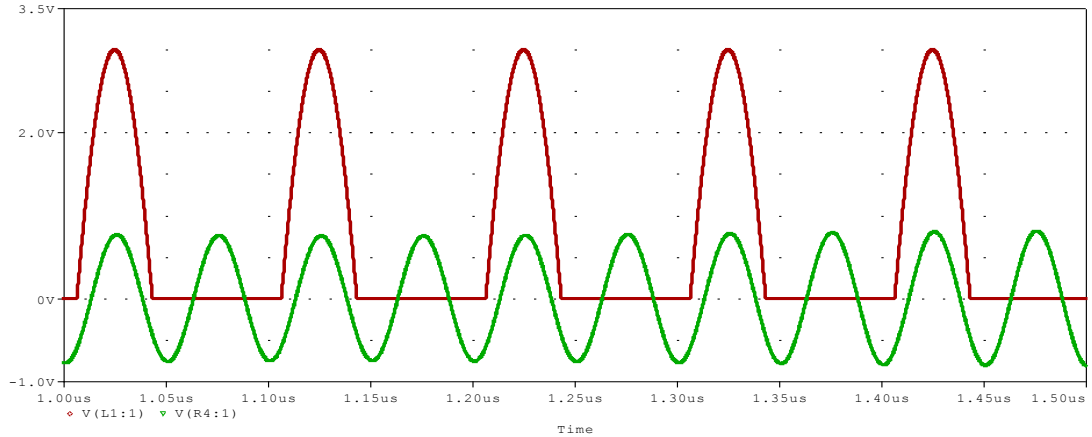

1.0V
0.5V
0V
0Hz
10MHz
20MHz
30MHz
40MHz
50MHz
60MHz
70MHz
80MHz
90MHz
100MHz
V(R3:2)
Frequency


3.5V
2.0V
0V
-1.0V
1.00us
1.05us
1.10us
1.15us
1.20us
1.25us
1.30us
1.35us
1.40us
1.45us
1.50us
V(R3:2)
V(L1:1)
Time


Vs
RL
0
0


1.5V
1.0V
0.5V
0V
0Hz
10MHz
20MHz
30MHz
40MHz
50MHz
60MHz
70MHz
80MHz
90MHz
100MHz
V(L1:1)
Frequency

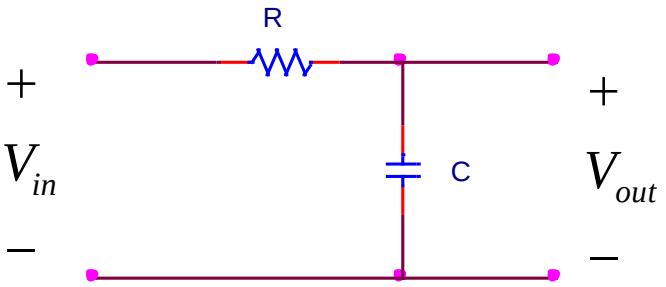

1.0V
0.5V
0V
0Hz
10MHz
20MHz
30MHz
40MHz
50MHz
60MHz
70MHz
80MHz
90MHz
100MHz
V(R4:1)
Frequency


3.5V
2.0V
0V
-1.0V
1.00us
1.05us
1.10us
1.15us
1.20us
1.25us
1.30us
1.35us
1.40us
1.45us
1.50us
V(L1:1)
V(R4:1)
Time

1. Mạch lọc thụ động

❖ Mạch lọc thông thấp, thông cao RC bậc 1

➢ Mạch lọc thông thấp RC bậc 1



$$H(S) = \frac{V_{out}(S)}{V_{in}(S)} = \frac{\frac{1}{SC}}{R + \frac{1}{SC}} \qquad \Rightarrow H(S) = \frac{1}{1 + SRC}$$

$$\Rightarrow H(j\omega) = \frac{1}{1 + j\omega.RC}$$

$$H(j\omega) = \frac{1}{1 + j\omega.RC}$$

$$H(j\omega) = |H(j\omega)|.e^{j.\varphi(\omega)}$$

- Module:

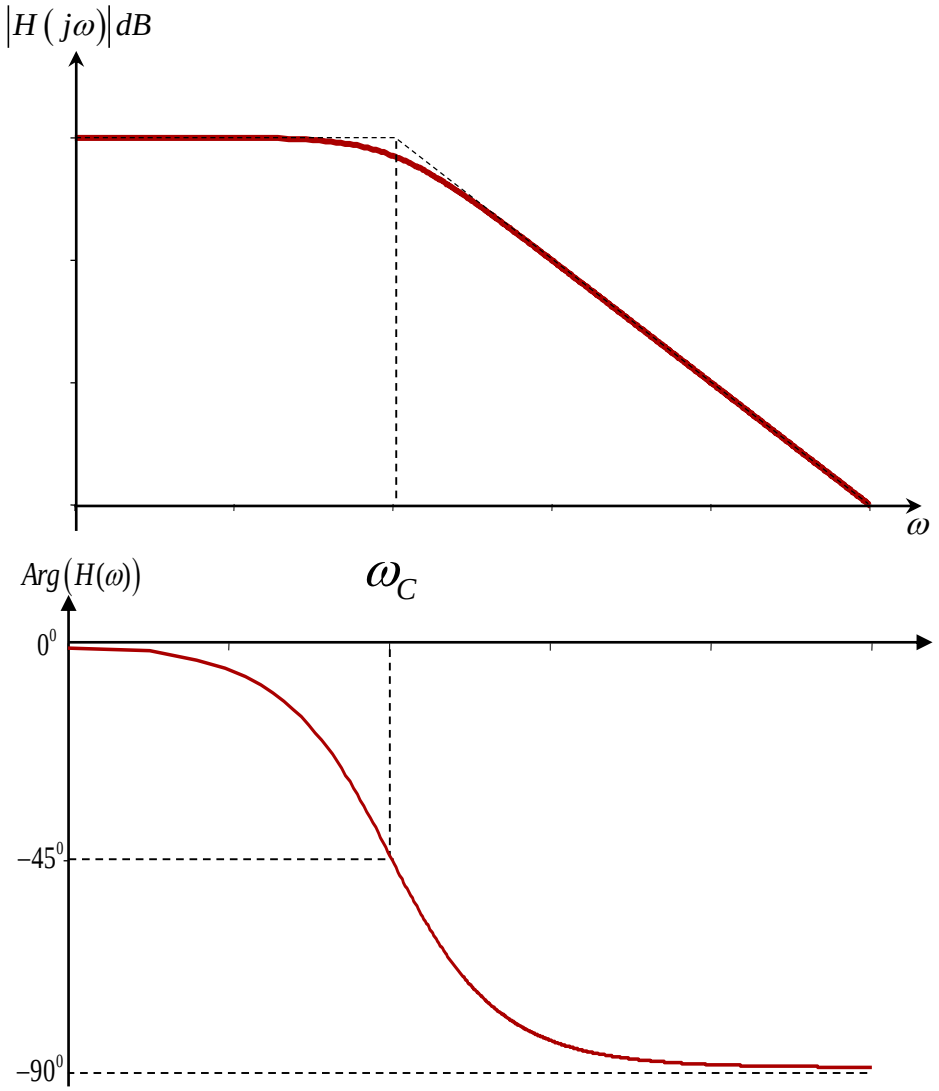
$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

- Argument:

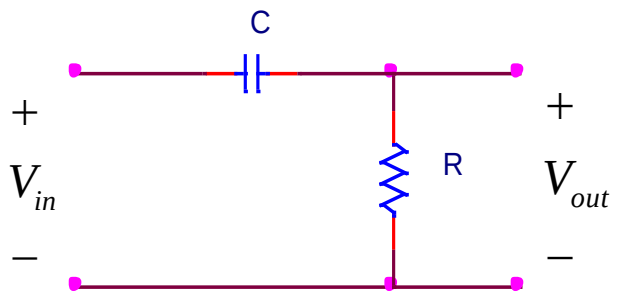
$$Arg\left(H(j\omega)\right) = -arctg(\omega RC)$$

- Tần số cắt:

$$\omega_C = \frac{1}{RC}$$



➢ Mạch lọc thông cao RC bậc 1



$$H(S) = \frac{V_{out}(S)}{V_{in}(S)} = \frac{R}{R + \frac{1}{SC}} \qquad \Rightarrow H(S) = \frac{SRC}{1 + SRC}$$

$$\Rightarrow H(j\omega) = \frac{j\omega.RC}{1 + j\omega.RC}$$

- Module:

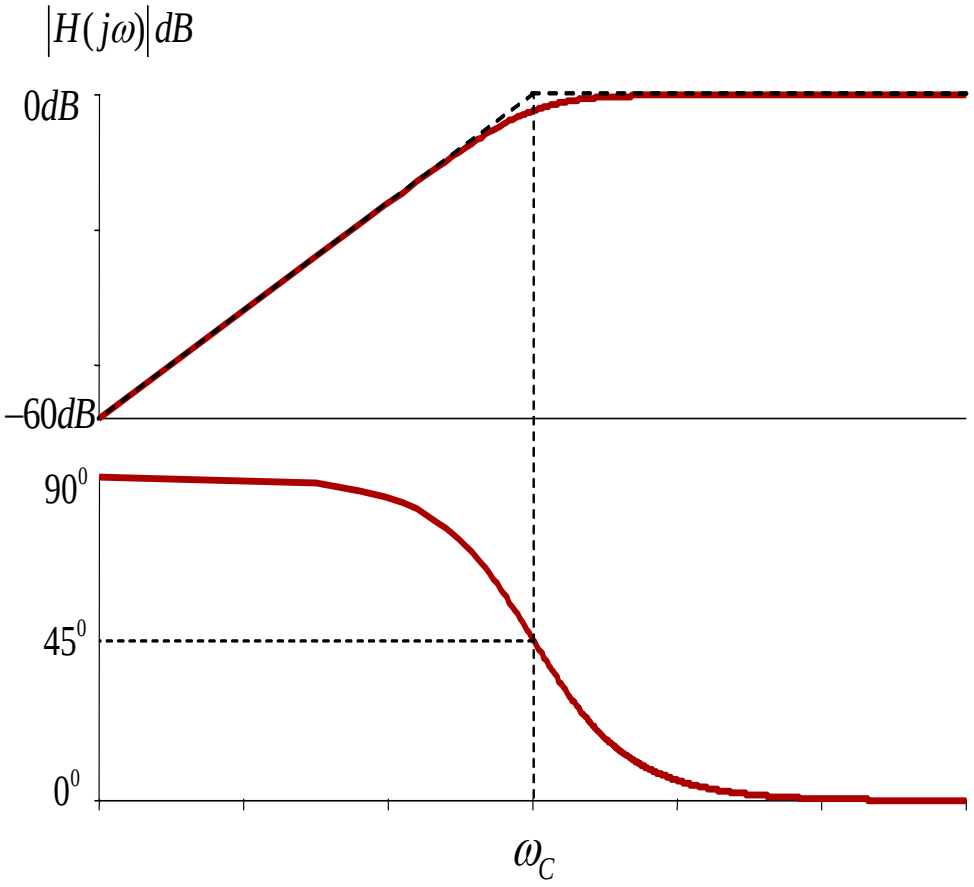
$$\left|H(j\omega)\right| = \frac{\omega RC}{\sqrt{1+\left(\omega RC\right)^2}}$$

- Argument:

$$Arg\left(H(j\omega)\right) = 90^0 - arctg(\omega RC)$$

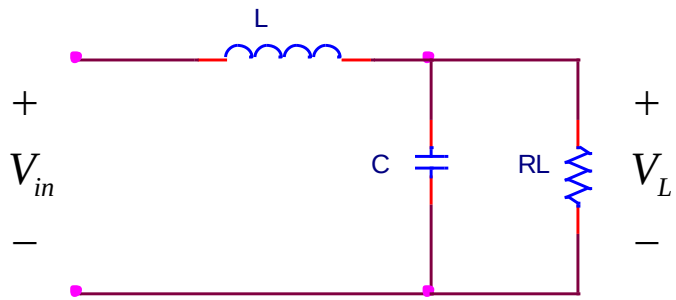
- Tần số cắt:

$$\omega_C = \frac{1}{RC}$$



❖ Mạch lọc RLC bậc 2

➢ Mạch lọc thông thấp LC



$$H(S) = \frac{V_L(S)}{V_{in}(S)} = \frac{\frac{1}{SC} // R_L}{SL + \frac{1}{SC} // R_L}$$

- $$\frac{1}{SC} // R_L = \frac{\frac{1}{SC}.R_L}{\frac{1}{SC} + R_L} = \frac{R_L}{1 + SR_L C}$$

$$\Rightarrow H(S) = \frac{\frac{R_L}{1 + SR_L C}}{SL + \frac{R_L}{1 + SR_L C}} = \frac{R_L}{S^2 R_L LC + SL + R_L} = \frac{\frac{1}{LC}}{S^2 + S\frac{1}{R_L C} + \frac{1}{LC}}$$

Đặt: $\omega_0^2 = \frac{1}{LC}$ $\quad 2\alpha = \frac{1}{R_L C}$ $\quad \Rightarrow H(S) = \frac{\omega_0^2}{S^2 + 2\alpha S + \omega_0^2}$

$$H(S)=\frac{\omega_0^2}{S^2+2\alpha S+\omega_0^2}=\frac{1}{\frac{S^2}{\omega_0^2}+\frac{2\alpha}{\omega_0^2}S+1}$$

Chuẩn hoá: $S_n=\frac{S}{\omega_0}$ $\Rightarrow H(S_n)=\frac{1}{S_n^2+2\alpha\frac{1}{\omega_0}S_n+1}$

$$\frac{2\alpha}{\omega_0}=\frac{1}{R_L C}\frac{1}{\omega_0}=\frac{1}{Q} \qquad Q=\frac{R_L}{\frac{1}{\omega_0 C}}$$

$$\Rightarrow H(S_n)=\frac{1}{S_n^2+\frac{1}{Q}S_n+1}$$

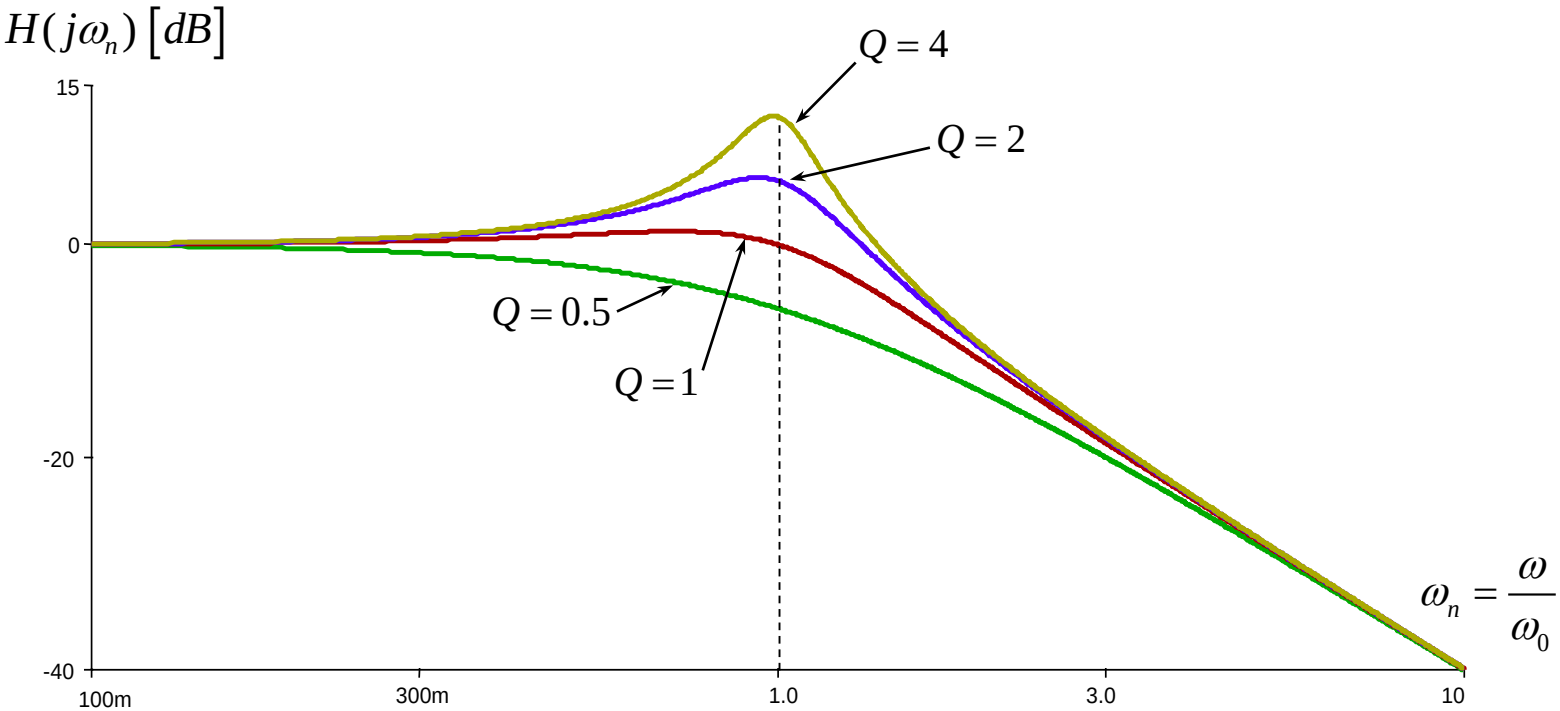
Nếu tín hiệu là tín hiệu điều hoà: $thay\ S_n=j\omega_n$ $\omega_n=\frac{\omega}{\omega_0}$

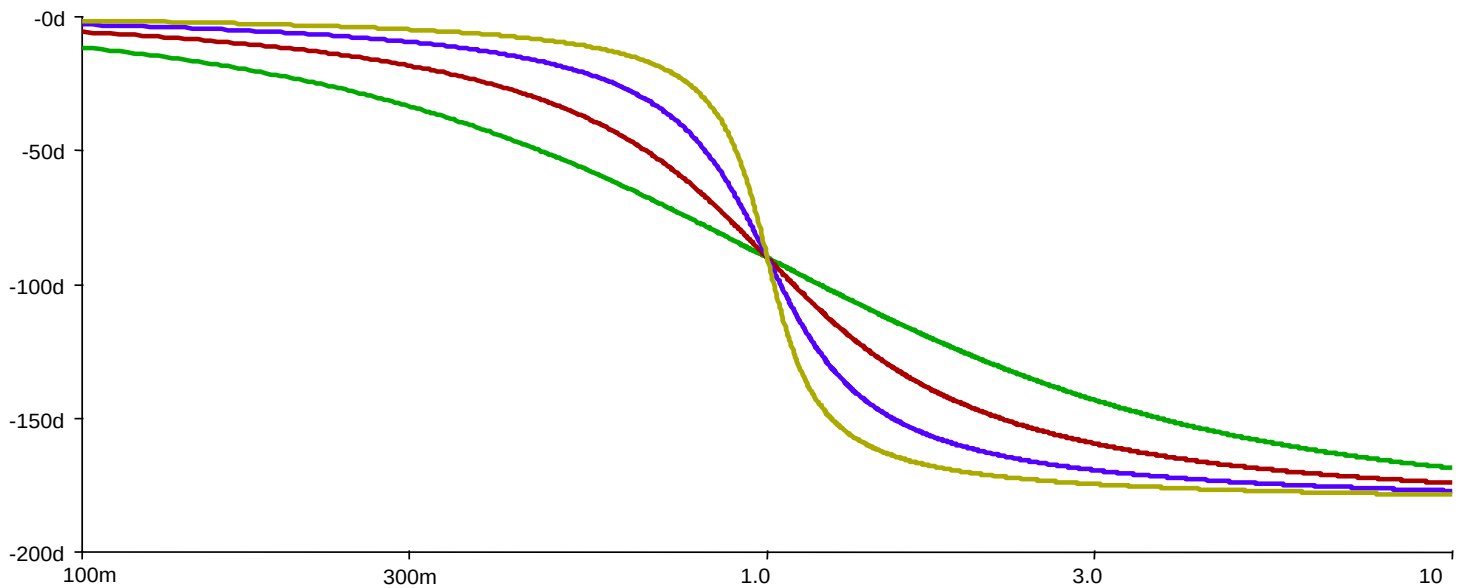
$$\Rightarrow H(j\omega_n)=\frac{1}{(j\omega_n)^2+j\frac{1}{Q}\omega_n+1}=\frac{1}{\left(1-\omega_n^2\right)+j\frac{1}{Q}\omega_n}$$

Biên độ: $$\left|H(j\omega_n)\right| = \frac{1}{\sqrt{\left(1-\omega_n^2\right)^2 + \left(\frac{1}{Q}\omega_n\right)^2}}$$

$$\left|H(j\omega_n)\right| dB = 20\log\left|H\left(j\omega_n\right)\right| = -20\log\sqrt{\left(1-\omega_n^2\right)^2 + \left(\frac{1}{Q}\omega_n\right)^2}$$

Pha: $$Arg\left[H(j\omega_n)\right] = -artg\,\frac{\frac{1}{Q}\omega_n}{1-\omega_n^2}$$


$H(j\omega_n)\,[dB]$
$Q = 4$
$Q = 2$
$Q = 0.5$
$Q = 1$
15
0
-20
-40
100m
300m
1.0
3.0
10
$\omega_n = \dfrac{\omega}{\omega_0}$


-0d
-50d
-100d
-150d
-200d
100m
300m
1.0
3.0
10

2. Mạch lọc Tích cực

❖ Mạch lọc có đáp ứng phẳng (Butteworth)

Đáp ứng về biên độ của mạch lọc Butteworth theo tần số chuẩn hoá:

$$\left|H(j\omega_n)\right| = \frac{1}{\sqrt{1+\left(\omega_n^k\right)^2}}$$

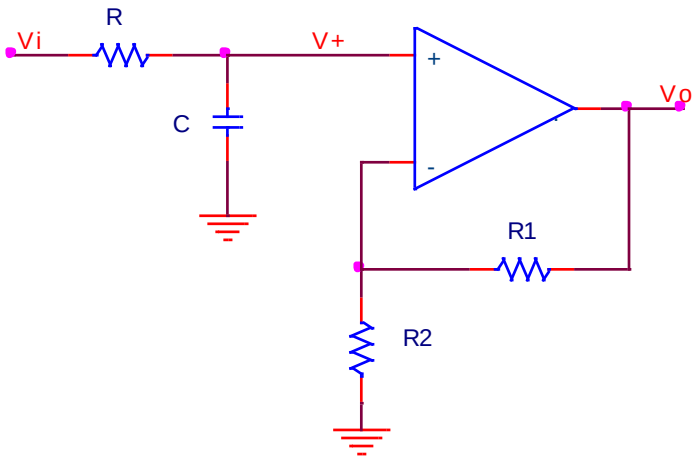
k là bậc của bộ lọc

Để thoả mãn điều này thì hàm truyền chuẩn hoá:

$$H(S_n) = \frac{1}{B_k(S_n)}$$

Bậc (k)	$B_k(S_n)$
1	$S_n + 1$
2	$S_n^2 + 1,414S_n + 1$
3	$S_n^3 + 2S_n^2 + 2S_n + 1 \;=\; (S_n+1).(S_n^2 + S_n + 1)$
4	$S_n^4 + 2,61S_n^3 + 3,41S_n^2 + 2,61S_n + 1 \;=\; (S_n^2 + 0,765S_n + 1).(S_n^2 + 1,848S_n + 1)$
5	$(S_n+1).(S_n^2 + 0,618S_n + 1).(S_n^2 + 1,616S_n + 1)$
6	$(S_n^2 + 0,518S_n + 1).(S_n^2 + 1,414S_n + 1).(S_n^2 + 1,932S_n + 1)$

➢Mạch lọc tích cực bậc 1



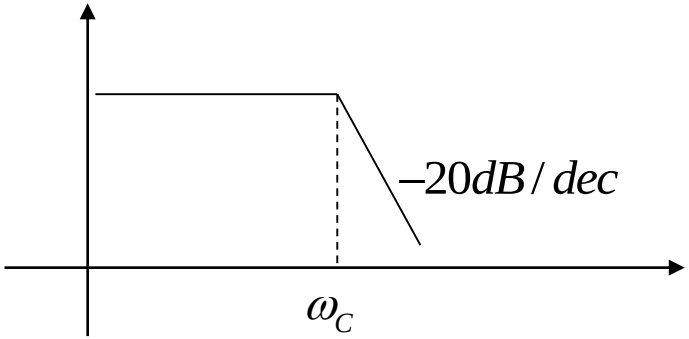
$$H(S)=\frac{V_0(S)}{V_i(S)}=\frac{V_0(S)}{V_+(S)}\cdot\frac{V_+(S)}{V_i(S)}=A_{V0}\cdot\frac{V_+(S)}{V_i(S)}$$

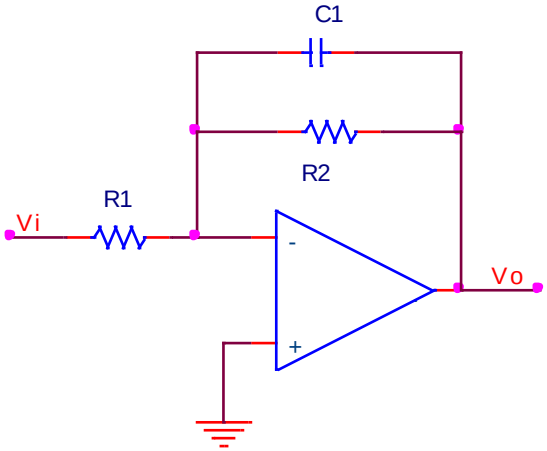
$$A_{V0}=\frac{R_1}{R_2}+1$$

$$\frac{V_+(S)}{V_i(S)}=\frac{V_C(S)}{V_i(S)}=\frac{\frac{1}{SC}}{R+\frac{1}{SC}}=\frac{1}{1+SRC}$$

Mạch lọc thông thấp

$$S_n=S.RC \qquad \Rightarrow H(S_n)=A_{V0}\cdot\frac{1}{S_n+1} \qquad \omega_C=\frac{1}{RC}$$

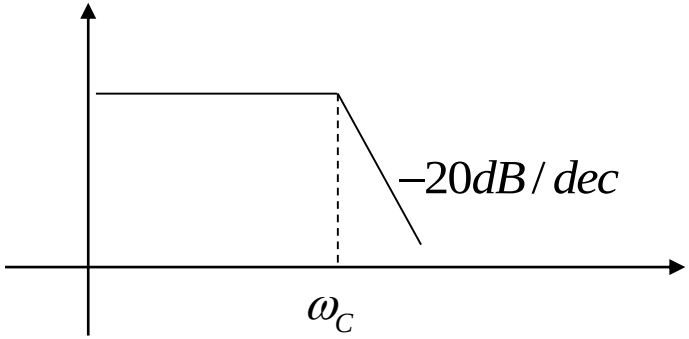


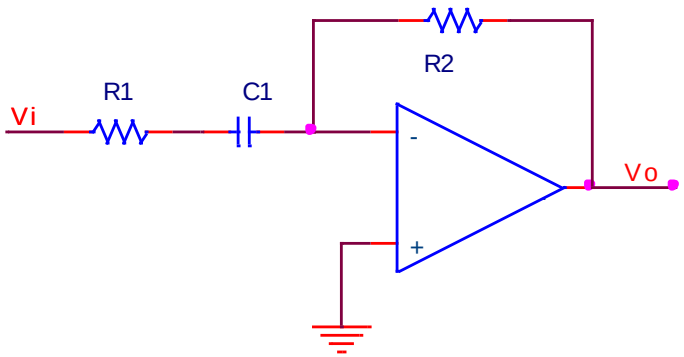


$$H(S)=\frac{V_0(S)}{V_i(S)}=-\frac{Z_f}{R_1}=-\frac{1}{R_1}.\frac{R_2}{1+SR_2C_1}=-\frac{R_2}{R_1}.\frac{1}{1+SR_2C_1}=\frac{A_{V0}}{1+SR_2C_1}$$

$$A_{V0}=-\frac{R_2}{R_1}\ ,\ \ \omega_C=\frac{1}{R_2C_1}$$

Mạch thông thấp 2

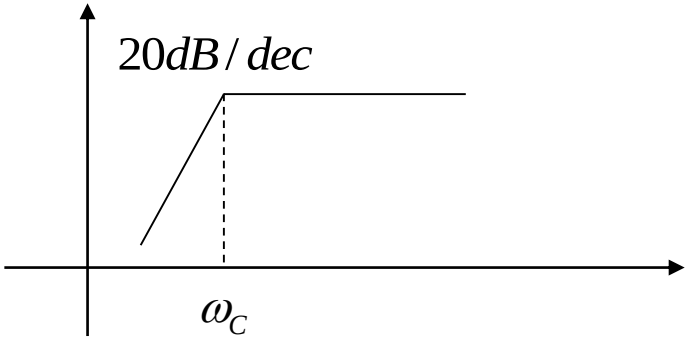




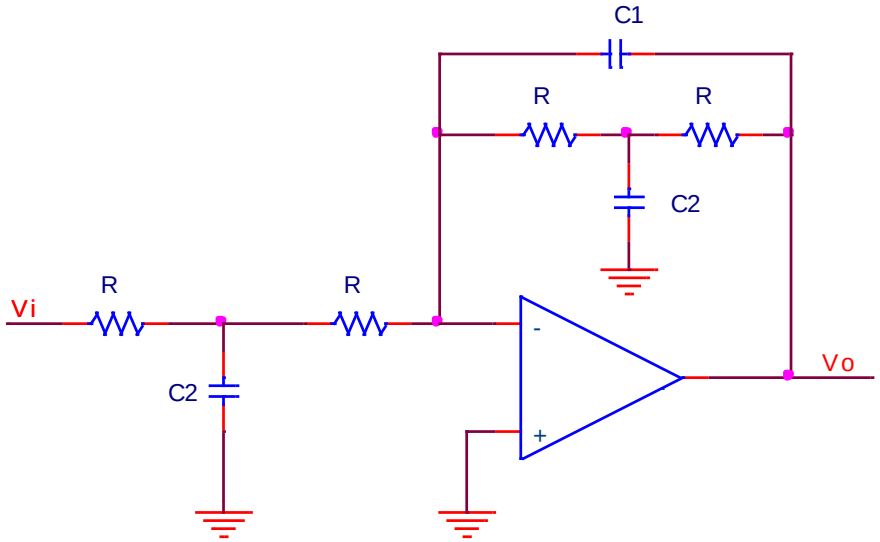
Mạch lọc thông cao

$$H(S)=\frac{V_0(S)}{V_i(S)}=-\frac{R_2}{Z_i}=-\frac{R_2}{R_1+1/SC_1}=-\frac{R_2}{R_1}.\frac{1}{1+\frac{1}{SR_1C_1}}=\frac{A_{V0}}{1+\frac{1}{SR_1C_1}}$$

$$S_n=S.R_1C_1 \qquad \Rightarrow H(S_n)=A_{V0}.\frac{1}{1+\frac{1}{S_n}} \qquad \omega_C=\frac{1}{R_1C_1}$$



➢Mạch lọc tích cực bậc 2

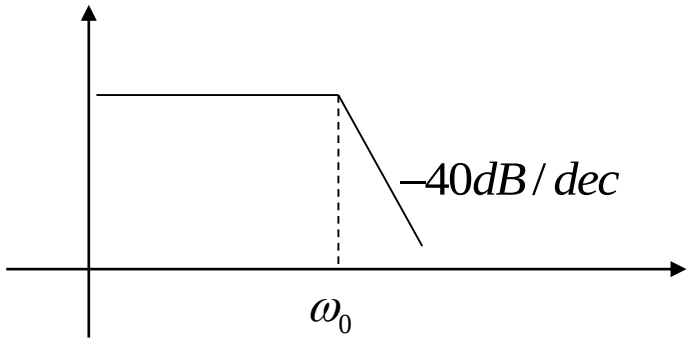


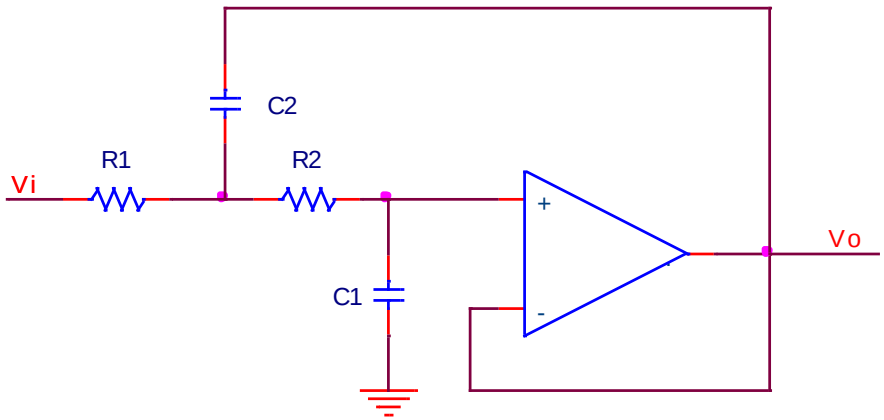
Mạch hồi tiếp âm 1 vòng

$$H(S)=\frac{V_0(S)}{V_i(S)}=\frac{1}{1+2\omega_0 RC_1 S+\omega_0^2 R^2 C_1 C_2 S^2}$$

$$\omega_0^2=\frac{1}{R^2 C_1 C_2}$$

$$\begin{cases} b_1=2\omega_0 RC_1=\sqrt{2} \\ b_2=\omega_0^2 R^2 C_1 C_2=1 \end{cases} \Rightarrow R, C_1, C_2$$





Mạch lọc thông thấp hồi tiếp dương

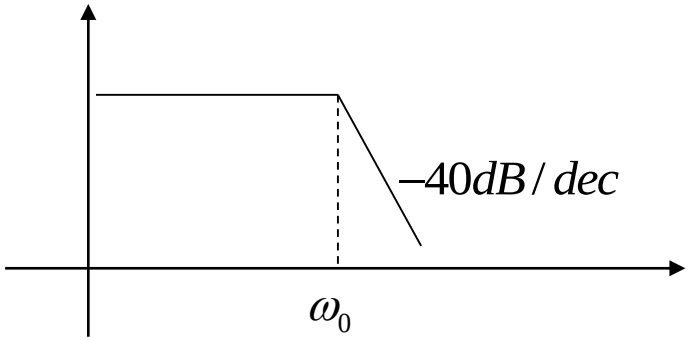
$$H(S) = \frac{V_0(S)}{V_i(S)} = \frac{1}{1 + C_1(R_1 + R_2)\omega_0 S + \omega_0^2 R_1 R_2 C_1 C_2 S^2}$$

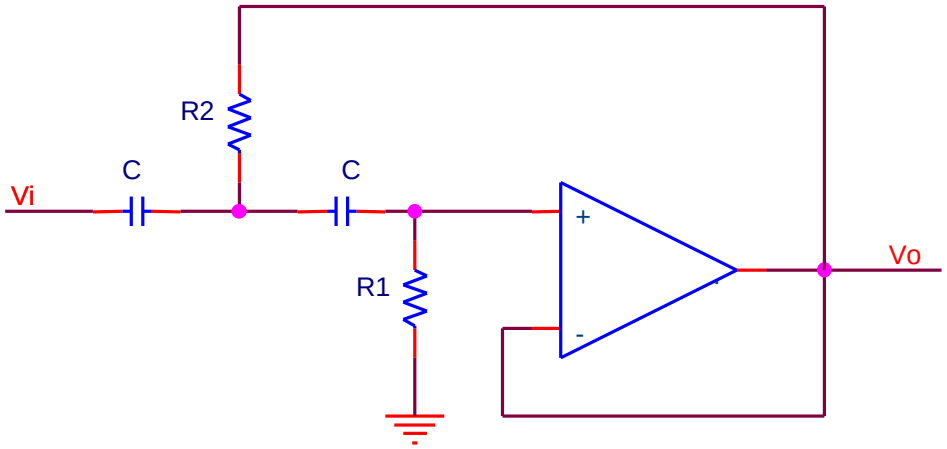
$$\omega_0^2 = \frac{1}{R_1 R_2 C_1 C_2}$$

$$\begin{cases} b_1 = \omega_0 C_1 (R_1 + R_2) = \sqrt{2} \\ b_2 = \omega_0^2 R_1 R_2 C_1 C_2 = 1 \end{cases}$$

Chọn: $\omega_0, C_1 \Rightarrow C_2$

$$\Rightarrow R_{1,2} = \frac{b_1 C_2 \pm \sqrt{b_1^2 C_2^2 - 4 b_2 C_1 C_2}}{4\pi f_0 C_1 C_2}$$





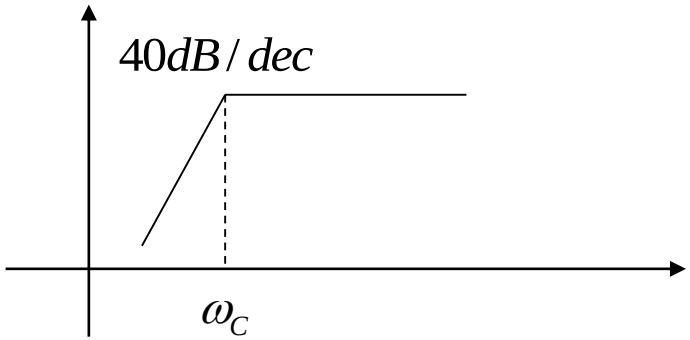
Mạch lọc thông cao hồi tiếp dương

$$H(S)=\frac{V_0(S)}{V_i(S)}=\frac{1}{1+\frac{2}{\omega_0 R_1 C}\cdot\frac{1}{S}+\frac{1}{\omega_0^2 R_1 R_2 C^2}\cdot\frac{1}{S^2}}$$

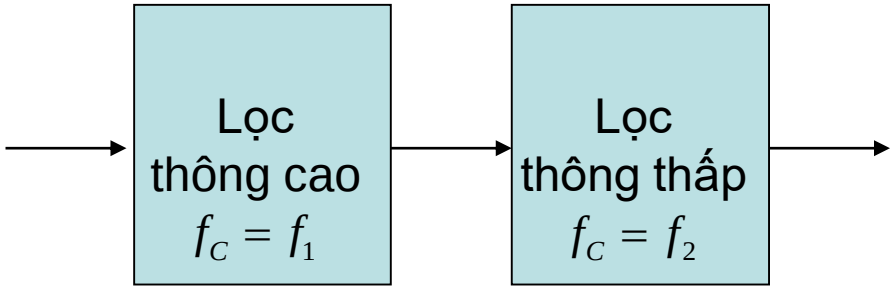
$$\omega_0^2=\frac{1}{R_1 R_2 C^2}$$

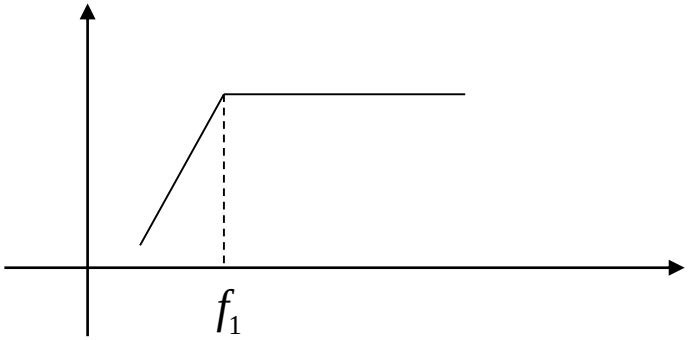
$$\begin{cases} b_1=\dfrac{2}{\omega_0 R_1 C}=\sqrt{2} \\ b_2=\dfrac{1}{\omega_0^2 R_1 R_2 C^2}=1 \end{cases}$$

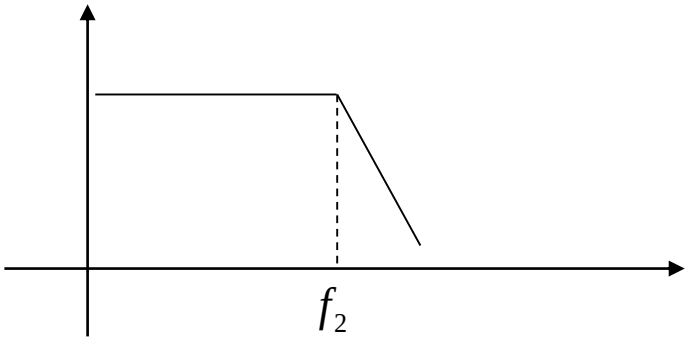
$$\Rightarrow R_1=\frac{\sqrt{2}}{\omega_0 C}\ ,\ \ R_1=\frac{R_1}{2}$$

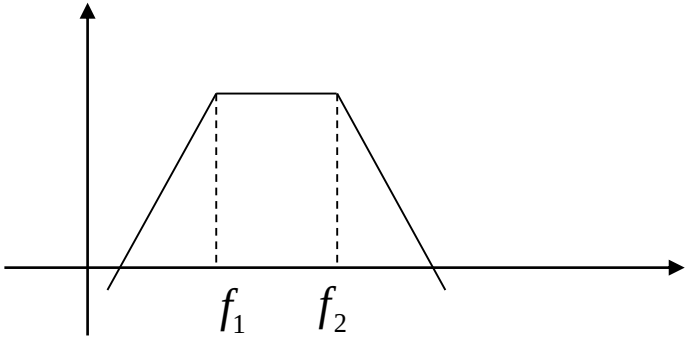


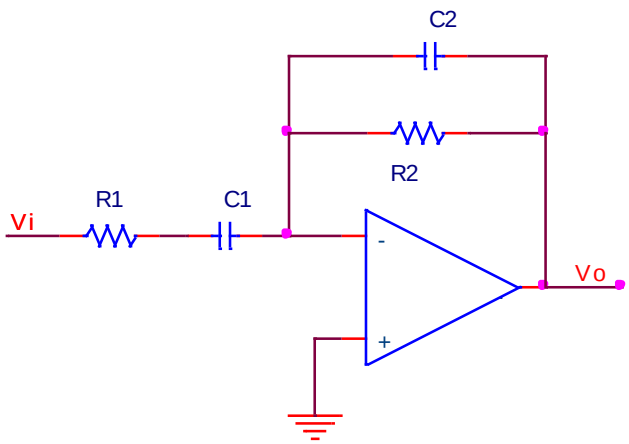
➢Mạch lọc Thông dải









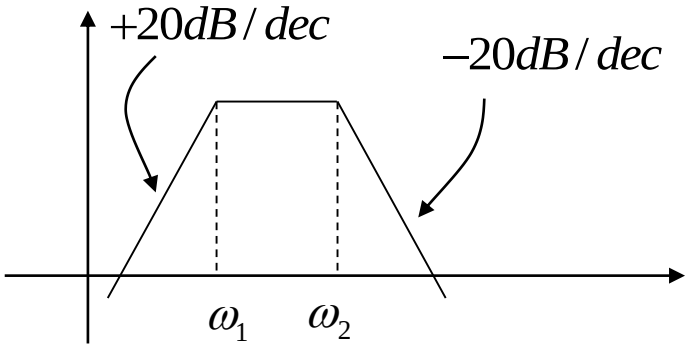


Mạch lọc thông dải bậc 2

$$H(S) = \frac{V_0(S)}{V_i(S)} = -\frac{Z_2}{Z_1}$$

$$Z_1 = R_1 + \frac{1}{SC_1} = \frac{1 + SR_1C_1}{SC_1}$$

$$Z_2 = \frac{R_2 \dfrac{1}{SC_2}}{R_2 + \dfrac{1}{SC_2}} = \frac{R_2}{1 + SR_2C_2}$$

$$\Rightarrow H(S) = \frac{-SR_2C_1}{(1 + S\,R_1C_1)(1 + S\,R_2C_2)}$$

$$\frac{1}{\omega_1} \qquad \frac{1}{\omega_2}$$